

FSH84211

## Citric acid monohydrate

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                                             |                                                                                                                                                                                                                                                  |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>产品说明:</b><br><b>Product Description:</b> | <b>柠檬酸</b><br><b>Citric acid monohydrate</b>                                                                                                                                                                                                     |
| <b>Cat No. :</b>                            | <b>A104-3; A104-3LC; A104-10; A104-10LC; A104-250LB; A104-500; A110-3; A110-10; A110-10LC; A110-50; A110-500; A111-12; A111-212; A111SAM-1; A111SAM-2; A111SAM-3; XXA00002KG; NC1675257</b><br>(Granular/Crystalline/Certified ACS/USP/EP/BP/JP) |
| <b>Synonyms</b>                             | 5949-29-1                                                                                                                                                                                                                                        |
| <b>CAS No</b>                               | C6 H8 O7 . H2 O                                                                                                                                                                                                                                  |
| <b>Molecular Formula</b>                    |                                                                                                                                                                                                                                                  |
| <b>Supplier</b>                             | Fisher Scientific Company<br>One Reagent Lane<br>Fair Lawn, NJ 07410<br>Tel: (201) 796-7100                                                                                                                                                      |
| <b>Emergency Telephone Number</b>           | CHEMTREC®, Inside the USA: 800-424-9300<br>CHEMTREC®, Outside the USA: 001-703-527-3887                                                                                                                                                          |
| <b>E-mail address</b>                       | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                   |
| <b>Recommended Use</b>                      | Laboratory chemicals.                                                                                                                                                                                                                            |
| <b>Uses advised against</b>                 | No Information available                                                                                                                                                                                                                         |

### SECTION 2. HAZARD IDENTIFICATION

|                                                                                                                                                |                            |                         |
|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------|
| <b>Physical State</b><br>Solid                                                                                                                 | <b>Appearance</b><br>White | <b>Odor</b><br>Odorless |
| <b>Emergency Overview</b><br>Causes serious eye irritation. May cause respiratory irritation. May form combustible dust concentrations in air. |                            |                         |

#### Classification of the substance or mixture

|                                                    |            |
|----------------------------------------------------|------------|
| Serious Eye Damage/Eye Irritation                  | Category 2 |
| Specific target organ toxicity - (single exposure) | Category 3 |

#### Label Elements


**Signal Word**
**Warning**

#### Hazard Statements

H319 - Causes serious eye irritation

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H335 - May cause respiratory irritation

**Precautionary Statements****Prevention**

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray

P264 - Wash face, hands and any exposed skin thoroughly after handling

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear eye protection/ face protection

**Response**

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P312 - Call a POISON CENTER or doctor if you feel unwell

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified. May form combustible dust concentrations in air.

**Health Hazards**

Causes serious eye irritation. May cause respiratory irritation.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. . Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

**Other Hazards**

May form explosible dust-air mixture if dispersed. This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component               | CAS No    | Weight % |
|-------------------------|-----------|----------|
| Citric acid monohydrate | 5949-29-1 | >95      |
| Citric acid             | 77-92-9   | -        |

**SECTION 4. FIRST AID MEASURES****General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Most important symptoms and effects**

None reasonably foreseeable.

**Self-Protection of the First Aider**

No special precautions required.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Dust can form an explosive mixture with air. Keep product and empty container away from heat and sources of ignition. Fine dust dispersed in air may ignite.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

**Exposure Controls**

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**Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment**

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |
| Butyl rubber   |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001; Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

**SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES**

|                                 |                               |                                          |
|---------------------------------|-------------------------------|------------------------------------------|
| <b>Appearance</b>               | White                         |                                          |
| <b>Physical State</b>           | Solid                         |                                          |
| <b>Odor</b>                     | Odorless                      |                                          |
| <b>Odor Threshold</b>           | No data available             |                                          |
| <b>pH</b>                       | 2.2                           | 50g/L (20°C)                             |
| <b>Melting Point/Range</b>      | 135 - 152 °C / 275 - 305.6 °F |                                          |
| <b>Softening Point</b>          | No data available             |                                          |
| <b>Boiling Point/Range</b>      | No information available      |                                          |
| <b>Flash Point</b>              | 173.9 °C / 345 °F             | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>         | Not applicable                | Solid                                    |
| <b>Flammability (solid,gas)</b> | No information available      |                                          |
| <b>Explosion Limits</b>         | No data available             |                                          |

|                                                |                                                                 |                                             |
|------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------|
| <b>Vapor Pressure</b>                          | No data available                                               |                                             |
| <b>Vapor Density</b>                           | Not applicable                                                  | Solid                                       |
| <b>Specific Gravity / Density</b>              | 1.54 g/cm <sup>3</sup> (20 °C)                                  |                                             |
| <b>Bulk Density</b>                            | 550 - 950 kg/m <sup>3</sup> (20 °C)                             |                                             |
| <b>Water Solubility</b>                        | 676 g/L (25°C)                                                  |                                             |
| <b>Solubility in other solvents</b>            | No information available                                        |                                             |
| <b>Partition Coefficient (n-octanol/water)</b> |                                                                 |                                             |
| <b>Component</b>                               | <b>log Pow</b>                                                  |                                             |
| Citric acid monohydrate                        | -1.72                                                           |                                             |
| Citric acid                                    | -1.72                                                           |                                             |
| <b>Autoignition Temperature</b>                | 345 °C / 653 °F                                                 |                                             |
| <b>Decomposition Temperature</b>               | > 170°C                                                         |                                             |
| <b>Viscosity</b>                               | Not applicable                                                  | Solid                                       |
| <b>Explosive Properties</b>                    |                                                                 | Dust can form an explosive mixture with air |
| <b>Oxidizing Properties</b>                    | Not oxidising                                                   |                                             |
| <b>Molecular Formula</b>                       | C <sub>6</sub> H <sub>8</sub> O <sub>7</sub> . H <sub>2</sub> O |                                             |
| <b>Molecular Weight</b>                        | 210.14                                                          |                                             |

## SECTION 10. STABILITY AND REACTIVITY

|                                         |                                                                                     |
|-----------------------------------------|-------------------------------------------------------------------------------------|
| <b>Stability</b>                        | Stable under normal conditions.                                                     |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                       |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                            |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat. Temperatures above 170°C. Avoid dust formation. |
| <b>Materials to avoid</b>               | Strong oxidizing agents. Strong bases.                                              |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ).                            |

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

## (a) acute toxicity;

| Component               | LD50 Oral             | LD50 Dermal     | LC50 Inhalation |
|-------------------------|-----------------------|-----------------|-----------------|
| Citric acid monohydrate | 5.79 g/kg ( Mouse )   |                 |                 |
| Citric acid             | LD50 = 3 g/kg ( Rat ) | >2 g/kg ( Rat ) |                 |

(b) skin corrosion/irritation; Based on available data, the classification criteria are not met

(c) serious eye damage/irritation; Category 2

## (d) respiratory or skin sensitization;

Respiratory  
SkinBased on available data, the classification criteria are not met  
Based on available data, the classification criteria are not met

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

(f) carcinogenicity; Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

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(g) reproductive toxicity; Based on available data, the classification criteria are not met

(h) STOT-single exposure; Category 3

Results / Target organs Respiratory system

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Target Organs None known.

(j) aspiration hazard; Not applicable  
Solid

Symptoms / effects, both acute and delayed No information available

## SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity effects Do not empty into drains. .

| Component   | Freshwater Fish                         | Water Flea          | Freshwater Algae | Microtox                                          |
|-------------|-----------------------------------------|---------------------|------------------|---------------------------------------------------|
| Citric acid | Leuciscus idus: LC50 = 440-760 mg/L/96h | EC50 = 120 mg/L/72h |                  | Photobacterium phosphoreum: EC50 = 14 mg/L/15 min |

Persistence and Degradability Readily biodegradable  
Persistence Persistence is unlikely.

Bioaccumulative Potential Bioaccumulation is unlikely

| Component               | log Pow | Bioconcentration factor (BCF) |
|-------------------------|---------|-------------------------------|
| Citric acid monohydrate | -1.72   | No data available             |
| Citric acid             | -1.72   | No data available             |

Mobility in soil The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors  
Persistent Organic Pollutant This product does not contain any known or suspected substance  
Ozone Depletion Potential This product does not contain any known or suspected substance

## SECTION 13. DISPOSAL CONSIDERATIONS

Waste from Residues/Unused Products Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

Contaminated Packaging Dispose of this container to hazardous or special waste collection point.

Other Information Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Solutions with low pH-value must be neutralized before discharge.

**SAFETY DATA SHEET**

Citric acid monohydrate

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport** Not Regulated**IMDG/IMO** Not regulated**IATA** Not regulated**Special Precautions for User** No special precautions required**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component               | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Citric acid monohydrate | -                                                   | -                                       | X    | X     | -         | -    | X   | X     | X    | X    | X    | -        |
| Citric acid             | -                                                   | -                                       | X    | X     | 201-069-1 | X    | X   | X     | X    | X    | X    | KE-20831 |

**National Regulations****SECTION 16. OTHER INFORMATION**

**Creation Date** 04-Sep-2009  
**Revision Date** 14-May-2024  
**Revision Summary** SDS sections updated.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Legend****CAS** - Chemical Abstracts Service**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**IECSC** - Chinese Inventory of Existing Chemical Substances**KECL** - Korean Existing and Evaluated Chemical Substances**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List**ENCS** - Japanese Existing and New Chemical Substances**AICS** - Australian Inventory of Chemical Substances**NZIoC** - New Zealand Inventory of Chemicals

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**WEL** - Workplace Exposure Limit**ACGIH** - American Conference of Governmental Industrial Hygienists**DNEL** - Derived No Effect Level**RPE** - Respiratory Protective Equipment**LC50** - Lethal Concentration 50%**NOEC** - No Observed Effect Concentration**PBT** - Persistent, Bioaccumulative, Toxic**TWA** - Time Weighted Average**IARC** - International Agency for Research on Cancer**PNEC** - Predicted No Effect Concentration**LD50** - Lethal Dose 50%**EC50** - Effective Concentration 50%**POW** - Partition coefficient Octanol:Water**vPvB** - very Persistent, very Bioaccumulative**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road**OECD** - Organisation for Economic Co-operation and Development**BCF** - Bioconcentration factor**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code**MARPOL** - International Convention for the Prevention of Pollution from Ships**ATE** - Acute Toxicity Estimate**VOC** - (Volatile Organic Compound)**Key literature references and sources for data**<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

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**End of Safety Data Sheet**